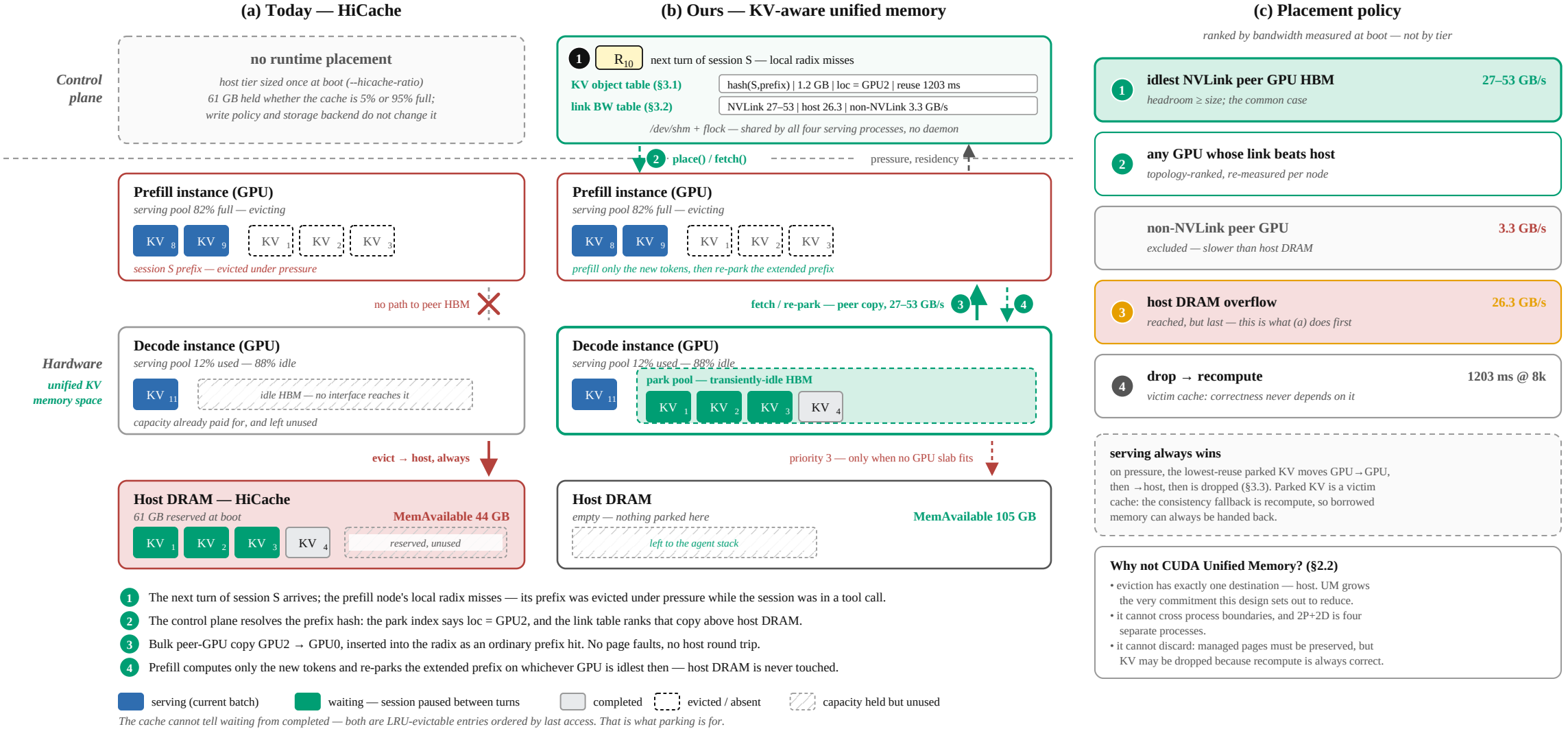
(a) Today — HiCache
(b) Ours — KV-aware unified memory
(c) Placement policy
ranked by bandwidth measured at boot — not by tier
Control plane
no runtime placement
host tier sized once at boot (--hicache-ratio)
61 GB held whether the cache is 5% or 95% full;
write policy and storage backend do not change it
1 R10 next turn of session S — local radix misses
KV object table (§3.1)
hash(S,prefix) | 1.2 GB | loc = GPU2 | reuse 1203 ms
link BW table (§3.2)
NVLink 27–53 | host 26.3 | non-NVLink 3.3 GB/s
/dev/shm + flock — shared by all four serving processes, no daemon
2 place() / fetch()
pressure, residency
Prefill instance (GPU)
serving pool 82% full — evicting
KV 8
KV 9
KV 1
KV 2
KV 3
session S prefix — evicted under pressure
prefill only the new tokens, then re-park the extended prefix
no path to peer HBM
fetch / re-park — peer copy, 27–53 GB/s
3
4
Hardware
unified KV memory space
Decode instance (GPU)
serving pool 12% used — 88% idle
KV 11
idle HBM — no interface reaches it
capacity already paid for, and left unused
park pool — transiently-idle HBM
KV 1
KV 2
KV 3
KV 4
evict → host, always
priority 3 — only when no GPU slab fits
Host DRAM — HiCache
61 GB reserved at boot
MemAvailable 44 GB
reserved, unused
Host DRAM
empty — nothing parked here
MemAvailable 105 GB
left to the agent stack
1 idlest NVLink peer GPU HBM 27–53 GB/s
headroom ≥ size; the common case
2 any GPU whose link beats host
topology-ranked, re-measured per node
non-NVLink peer GPU 3.3 GB/s
excluded — slower than host DRAM
3 host DRAM overflow 26.3 GB/s
reached, but last — this is what (a) does first
4 drop → recompute 1203 ms @ 8k
victim cache: correctness never depends on it
serving always wins
on pressure, the lowest-reuse parked KV moves GPU→GPU, then →host, then is dropped (§3.3). Parked KV is a victim cache: the consistency fallback is recompute, so borrowed memory can always be handed back.
Why not CUDA Unified Memory? (§2.2)
• eviction has exactly one destination — host. UM grows the very commitment this design sets out to reduce.
• it cannot cross process boundaries, and 2P+2D is four separate processes.
• it cannot discard: managed pages must be preserved, but KV may be dropped because recompute is always correct.
1 The next turn of session S arrives; the prefill node's local radix misses — its prefix was evicted under pressure while the session was in a tool call.
2 The control plane resolves the prefix hash: the park index says loc = GPU2, and the link table ranks that copy above host DRAM.
3 Bulk peer-GPU copy GPU2 → GPU0, inserted into the radix as an ordinary prefix hit. No page faults, no host round trip.
4 Prefill computes only the new tokens and re-parks the extended prefix on whichever GPU is idlest then — host DRAM is never touched.
serving (current batch)
waiting — session paused between turns
completed
evicted / absent
capacity held but unused
The cache cannot tell waiting from completed — both are LRU-evictable entries ordered by last access. That is what parking is for.